Project 01 - Simplex

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1 History of the Simplex Method

- George B. Dantzig developed the Simplex method in 1947 to solve linear programming problems.
- The method seeks an optimal solution among the basic feasible solutions, associated with the vertices of the feasible region.
- In each iteration, a variable enters the base and another leaves, resulting in a new basic feasible solution. This is known as pivot operation.
- In bounded problems, iterations advance toward an improvement of the objective; in degenerate cases, the value may remain the same.
- It can detect optimality, unboundedness, and, according to the formulation, infeasibility.
- It is efficient in many practical problems, although its worst-case theoretical complexity is exponential.

Problem

Name: Linear Programming Problem
Course:
Members:
Semester:

Mathematical Model

$$\max \quad Z = 7\,x_1 + 2\,x_2$$

$$\begin{aligned} \text{S.t.} \; 2\,x_1 + 2\,x_2 &\leq 9 \\ 3\,x_1 + 1\,x_2 &\leq 13 \\ x_j &\geq 0 \quad \text{for every variable } j \end{aligned}$$

2 Initial table

Row	Z	x_1	x_2	S_1	S_2	-
Z	1	-7	-2	0	0	0
R_1	0	2	2	1	0	9
R_2	0	3	1	0	1	13

Middle tables

Table states were saved during the iterations

3 After step 1

Row	Z	x_1	x_2	S_1	S_2	-
Z	1	0	0,3333	0	2,333	30,33
R_1	0	0	1,333	1	-0,6667	0,3333
R_2	0	1	0,3333	0	0,3333	4,333

Result

Optimal solution found.

4 Final Table

Row	Z	x_1	x_2	S_1	S_2	-
Z	1	0	0,3333	0	2,333	30,33
R_1	0	0	1,333	1	-0,6667	0,3333
R_2	0	1	0,3333	0	0,3333	4,333

5 Problem Solution

- An optimal solution was found.
- **Optimal Solution**
  - Objective Function Value: $Z = 30,3333$.
  - `x1` $= 4,33333$ (basic)
  - `x2` $= 0$ (non-basic)
  - $S_1 = 0,333333$ (basic)
  - $S_2 = 0$ (non-basic)